D02 · Capacity & Long-Range Planning

01 · Rough-Cut Capacity Planning (RCCP)

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1 WHAT IT IS

RCCP is the feasibility check that sits between the demand consensus and the detailed MRP run — a validation of whether the Master Production Schedule (MPS) is producible against the capacity of a handful of critical resources: bottleneck work centers, key shared cells, long-lead-time materials, and constrained skilled labor pools.

2 WHY IT MATTERS

It is the gate where the plan becomes credible — or doesn't. **The MPS that passes RCCP is a commitment the plant can keep**; the one that skips it is a forecast disguised as a plan. The dominant failure pattern is well-documented in capacity-constrained operations **(Vollmann, Berry, Whybark & Jacobs, 2005 — textbook of record)**: schedules built on demand-side optimism collide with bottleneck reality in week three of the quarter, surfacing as missed deliveries that cost more to recover than they would have cost to prevent. The asymmetry is structural — **capacity is committed in months but exposed in weeks**, so an early signal is more valuable than a precise one.

3 HOW IT IS DONE

1. **Build the Bill of Resources (BoR) before anything else.** Map each product family to the critical resources it consumes — minutes per unit on each bottleneck, energy, operator-hours, changeover penalty per transition. Without it, the model is a guess.
2. **Select 8–15 critical resources — no more.** Use Theory of Constraints to identify bottlenecks, shared cells, long-LT imports and scarce skills. A bigger model is not a better one; it is a slower, noisier approximation of CRP.
3. **Compute load by family, not SKU.** RCCP is directional, not detailed. Aggregate to the level that captures meaningful constraint signal without forcing the planner into SKU-level noise.
4. **Emit one of three signals: feasible, constrained, infeasible.** A constrained signal demands quantified mitigation options (overtime, weekend shifts, co-manufacturing, alt routing). An infeasible signal sends the plan back to the demand review.
5. **Convert every flagged constraint into pre-S&OP options** with cost, time and risk attached. The executive S&OP does not discover constraints — it decides among options the pre-S&OP has already shaped.
6. **Track predictive accuracy and re-calibrate the BoR semi-annually.** Every RCCP signal is a forecast. Compare it to actual execution; feed misses back into BoR coefficients. Without this loop, the model decays silently into background noise.

IN PRACTICE

Keep the critical-resource set tight

Restrict RCCP to the handful of resources that actually constrain the plan, and review the set as bottlenecks shift. A bloated resource list slows the check and dilutes attention; the power of rough-cut is that it looks only where infeasibility actually originates.

Generate quantified options per flagged constraint

When RCCP flags an overload, hand the supply review a set of quantified options — overtime, alternate routing, pull-ahead, deferral — with cost and feasibility, not just a red flag. A constraint surfaced without options is an alarm; a constraint surfaced with options is a decision.

Measure the model's predictive accuracy

Check quarterly whether the loads RCCP predicted matched what the critical resources actually experienced. A Bill of Resources drifts as routings and rates change; an RCCP model that is never validated against reality slowly becomes a confident, wrong feasibility check.

Position RCCP inside the S&OP supply review

RCCP is the analytical engine of the supply-balancing step: it is where the demand plan meets finite critical capacity. Wiring it into the S&OP supply review makes the feasibility check a governed gate rather than an optional analysis a planner runs when there is time.

WHAT YOU WOULD USE

RCCP / CAPACITY VALIDATION

SAP IBP / S4 (RCCP) / o9 / Kinaxis — enterprise tier-1 — Native rough-cut and detailed capacity checks integrated with the MPS and the S&OP supply review, with Bill of Resources modeling. **(SAP, o9, Kinaxis — vendor)** The choice when the feasibility gate must sit inside the integrated plan.

Blue Yonder / Oracle — capacity planning — Solid RCCP/CRP modules with resource-load visualization and option modeling. **(Blue Yonder, Oracle — vendor)** A strong fit where the MPS-to-capacity check is the priority within an existing APS/ERP footprint.

Finite-capacity / load-leveling specialists — Dedicated capacity-modeling and load-leveling tools for complex resource networks. **(specialized vendors — vendor)** The vertical option when the critical-resource topology is intricate enough to exceed packaged RCCP.

For operations that don't need a platform — A Bill of Resources for the handful of critical resources and a load-vs-capacity check in a workbook, run at the MPS level each S&OP cycle. The discipline of testing the schedule against critical capacity before committing — not the tool — is what converts the MPS from a wish into a plan.

4 SO WHAT

The MPS that has not passed RCCP is not a plan — it is a wish. Own the Bill of Resources, keep the critical-resource set tight, generate quantified options for every flagged constraint, and measure the model's predictive accuracy quarterly. Capacity is committed in months but exposed in weeks; RCCP is the only mechanism that closes that asymmetry before customers do.

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